

Targeted learning with right-censored data using the state learner

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Targeted or debiased machine learning*

Low-dimensional target parameter

$$\Psi: \mathcal{P} \rightarrow \mathbb{R}, \quad \text{such that} \quad \Psi(P) = \tilde{\Psi}(\nu_1(P), \dots, \nu_B(P)),$$

for high-dimensional nuisance parameters $\nu_1, \dots, \nu_B$.

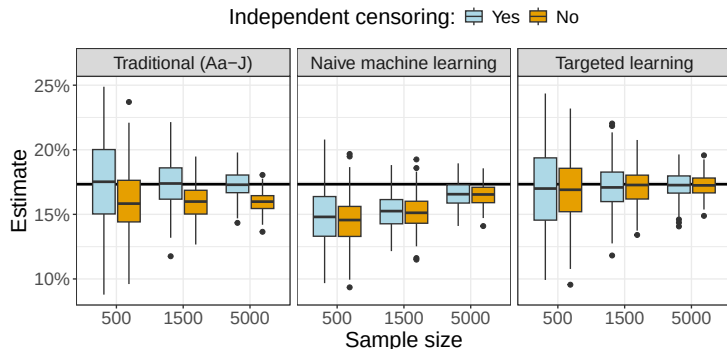
*van der Laan and Rose [2011], Chernozhukov et al. [2018]

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Adapted from Figure 4 in [Munch et al., 2023].

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Examples of estimands

Available data $O = (X, \tilde{T}, \tilde{D}) \sim P \in \mathcal{P}$, where

$$X = W \in \mathbb{R}^d \text{ or } X = (W, A) \in \mathbb{R}^d \times \{0, 1\},$$

$$\tilde{T} = T \wedge C$$

$$\tilde{D} = \mathbb{1}\{T \leq C\}D, \quad D = 1 \text{ or } D \in \{1, 2\}.$$

The variables T and D are the uncensored event time and event indicator.

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$$Q(T > t) \stackrel{(!)}{=} \mathbb{E}_P \left[\prod_0^t (1 - \Lambda_P(ds | X)) \right], \quad (T, X) \sim Q,$$

where

$$\Lambda_P(dt | x) = P(\tilde{T} \in dt, \tilde{D} = 1 | \tilde{T} \geq t, X = x).$$

For (!) to hold we need $C \perp\!\!\!\perp T | X$.

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Use $\{(T^a, D^a) : a \in \{0, 1\}\} \sim Q$, to denote potential outcomes.

$$Q(T^a \leq t, D^a = 1) \stackrel{(!)}{=} \mathbb{E}_P \left[\int_0^t \prod_0^u (1 - [\Lambda_{1,P} + \Lambda_{2,P}](ds \mid a, W)) \Lambda_{1,P}(du \mid a, W) \right],$$

$$\Lambda_{d,P}(dt \mid a, w) = P(\tilde{T} \in dt, \tilde{D} = d \mid \tilde{T} \geq t, A = a, W = w).$$

For (!) to hold we need both $C \perp\!\!\!\perp T \mid X$ and causal assumptions.

Targeted learning with super learning

$$\Psi(P) = \mathbb{E}_P \left[\int_0^t \prod_0^u (1 - [\Lambda_{1,P} + \Lambda_{2,P}](ds \mid a, W)) \Lambda_{1,P}(du \mid a, W) \right]$$

can be written as

$$\Psi(P) = \tilde{\Psi}(\Lambda_{1,P}, \Lambda_{2,P}, P_W).$$

$$\tilde{\Psi}(\hat{\Lambda}_{1,n}, \hat{\Lambda}_{2,n}, \mathbb{P}_n) \quad + \quad \begin{array}{c} \text{Targeting/debiasing step} \\ \text{using } \hat{\Gamma}_n \text{ and } \hat{\pi}_n \end{array} \longrightarrow \hat{\Psi}_n^*$$

Asymptotic valid inference and $O_P(n^{-1/2})$ convergence for $\hat{\Psi}_n^*$ if

$$\|\hat{\nu}_n - \nu\|_{P,2} = o_P(n^{-1/4}), \quad \text{for all } \hat{\nu}_n \in \{\hat{\Lambda}_{1,n}, \hat{\Lambda}_{2,n}, \hat{\Gamma}_n, \hat{\pi}_n\}.$$

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Attempt to achieve this using a super learner.

Super learning[†]

$$\mathcal{D}_n = \begin{array}{|c|c|c|c|c|} \hline O_1 & O_2 & \dots & \dots & \dots \\ \hline \end{array}$$

Learner $\mathcal{D}_n \mapsto a(\mathcal{D}_n) = \hat{\nu}_n$

Library $\mathcal{A} = \{a_1, a_2, \dots, a_M\}$

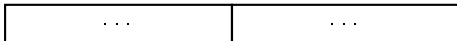
Loss function $\mathcal{V} \times \mathcal{O} \ni (\nu, O) \mapsto L(\nu, O) \in \mathbb{R}$

$$\text{Discrete SL} = \hat{a}_n = \operatorname{argmin}_{a \in \mathcal{A}} \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathcal{D}_n^k|} \sum_{O_i \in \mathcal{D}_n^k} L(a(\mathcal{D}_n^{-k}), O_i),$$

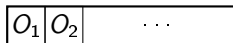
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Super learning[†]

$$\mathcal{D}_n^{-1} =$$



$$\mathcal{D}_n^1 =$$



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Super learning[†]

$$\mathcal{D}_n^{-2} = \begin{array}{|c|c|c|} \hline O_1 & O_2 & \dots \\ \hline \end{array} \qquad \begin{array}{|c|} \hline \dots \\ \hline \end{array}$$
$$\mathcal{D}_n^2 = \begin{array}{|c|} \hline \dots \\ \hline \end{array}$$

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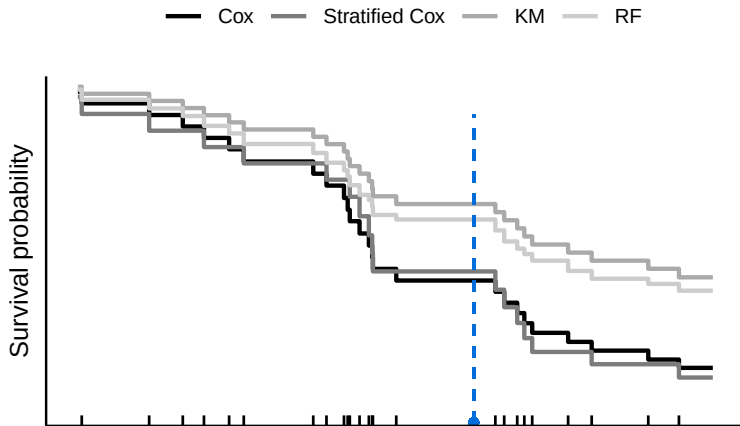
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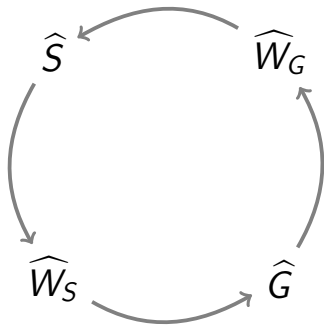
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Evaluating performance in hold-out folds is a challenge with right-censored data

The negative log-likelihood is unsuited for super learning

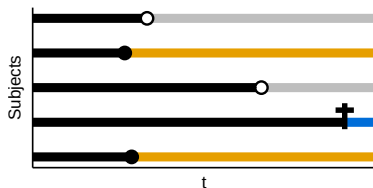


IPCW and pseudo-outcome require pre-specified censoring estimator



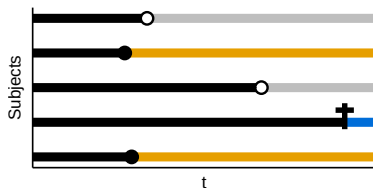
The state learner: Model all 'states' of the *observed* data

$$N(t) = \mathbb{1}\{\tilde{T} \leq t, \tilde{D} = 1\} + 2\mathbb{1}\{\tilde{T} \leq t, \tilde{D} = 2\} - \mathbb{1}\{\tilde{T} \leq t, \tilde{D} = 0\}$$



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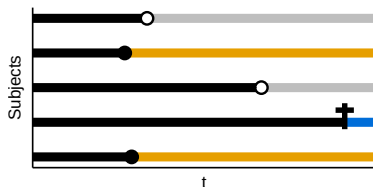


Build a super learner for the function

$$F(t, k, x) = P(N(t) = k \mid X = x).$$

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Build a super learner for the function

$$F(t, k, x) = P(N(t) = k \mid X = x).$$

$$L(F, O) = \int_0^\tau \sum_{j=-1}^2 \left(F(t, j, X) - \mathbb{1}\{N(t) = j\} \right)^2 dt.$$

Building a library for F

The formulas

$$F(t, 0, x) = \prod_0^t (1 - [\Lambda_1 + \Lambda_2 + \Gamma](ds \mid x)),$$

$$F(t, j, x) = \int_0^t F(t-, 0, x) \Lambda_j(ds \mid x), \quad j \in \{1, 2\},$$

$$F(t, -1, x) = \int_0^t F(t-, 0, x) \Gamma(ds \mid x),$$

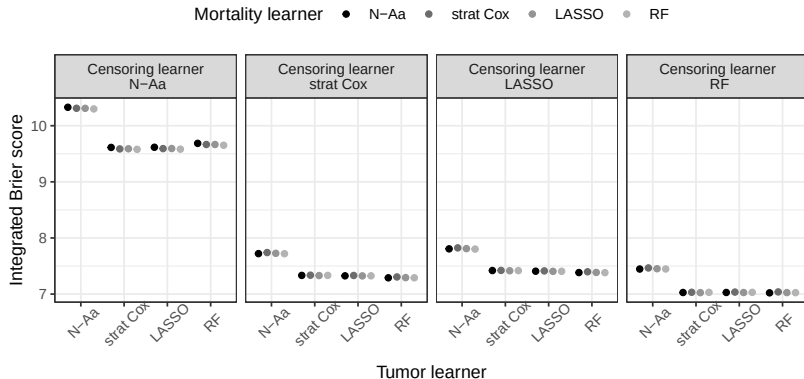
can be used to build a library for F from libraries for Λ_1 , Λ_2 , Γ ,

$$\mathcal{F}(\mathcal{A}, \mathcal{B}, \mathcal{C}) = \{F_{a,b,c} : a \in \mathcal{A}, b \in \mathcal{B}, c \in \mathcal{C}\}.$$

Similarly, we can obtain Λ_j and Γ from F .

Illustration with data from a prostate cancer study[‡]

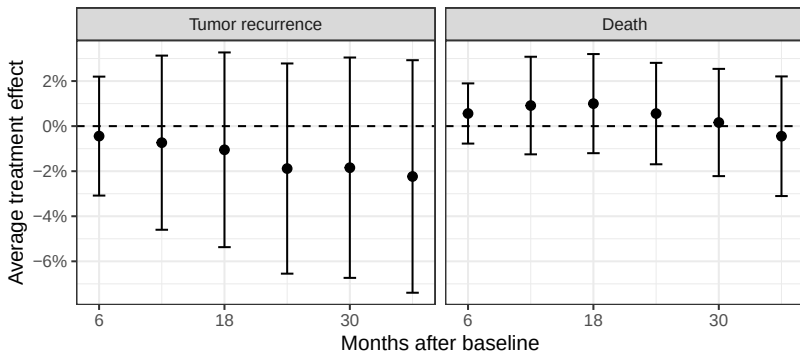
The learners selected by the state learner.



[‡]Kattan et al. [2000].

Illustration with data from a prostate cancer study[‡]

The target parameter was the average treatment effect of hormone therapy.



[‡]Kattan et al. [2000].

Additional details

Munch and Gerds [2024] provide

- a consistency result
- a finite sample oracle inequality
- details for the targeting step

Prototype available at <https://github.com/amnudn/statelearner>.

Thank you for the attention!

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